

Class 05: Data Visualization with GGLOT

Joli DeLucia (A17320476)

Table of contents

Background	1
Gene Expression Plot	4
Going Further with gapminder	29
First look at the dplyr package	32

Background

There are lot's of ways to make plots in R. These include so-called “base R” (like the `plot()`) and add on packages like **ggplot2**.

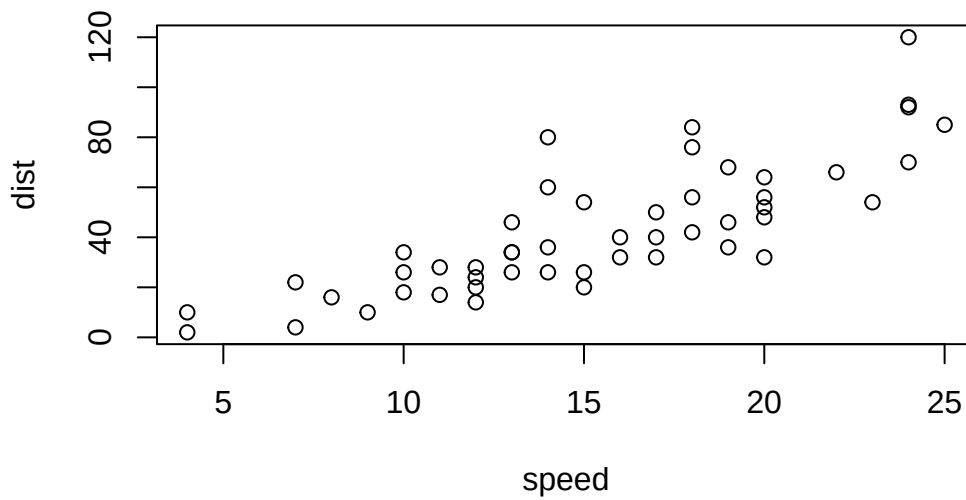
Let's make the same plot with these two graphics systems. We can use the inbuilt `cars` data set:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With “base R” we can simply:

```
plot(cars)
```

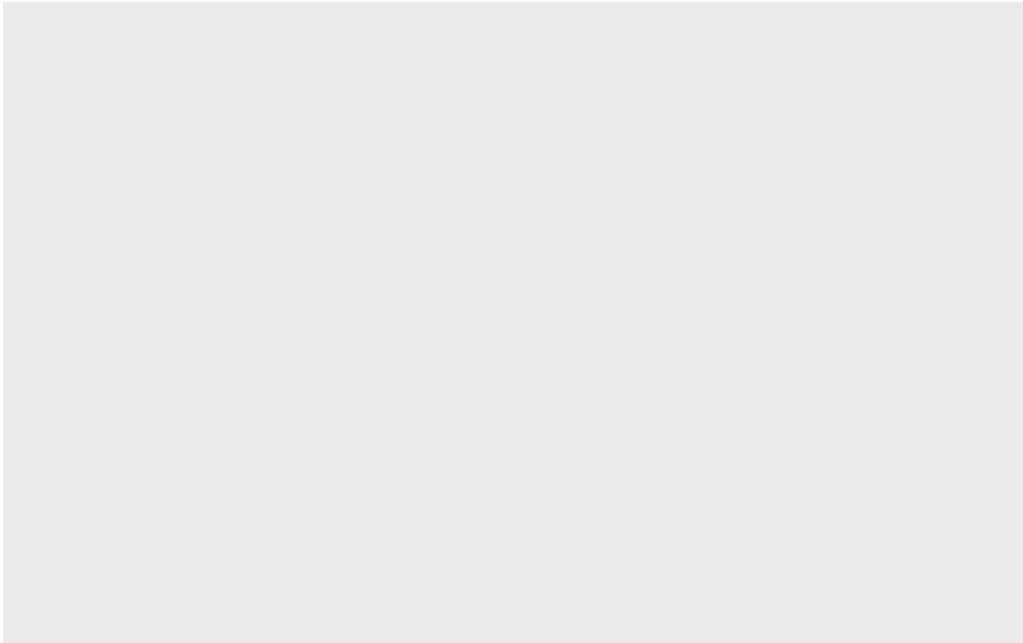


Now let's try ggplot. First I need to install the package using `install.packages("ggplot2")`

N.B. We never run an `install.packages()` in a code chunk otherwise we will re-install needlessly every time we render our document.

Everytime you want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

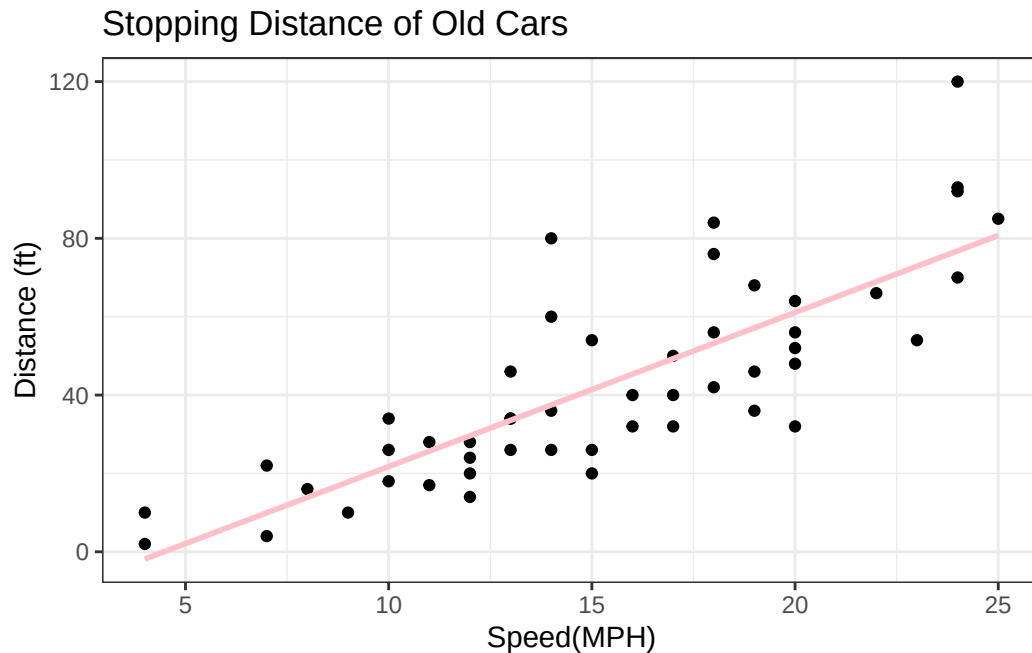


Every ggplot needs at least 3 things:

1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** or aesthetics that map the data to the plot
3. The **geom** or geometry i.e the plot type such as points, lines, etc.

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()+  
  geom_smooth(method=lm, se = FALSE, color = "pink") +  
  labs(x = "Speed(MPH)", y = "Distance (ft)", title = "Stopping Distance of Old Cars") +  
  theme_bw()
```

``geom_smooth()`` using formula = 'y ~ x'



Gene Expression Plot

Read some data on the effect of GLP-1 inhibitor (drug) on gene expression values:

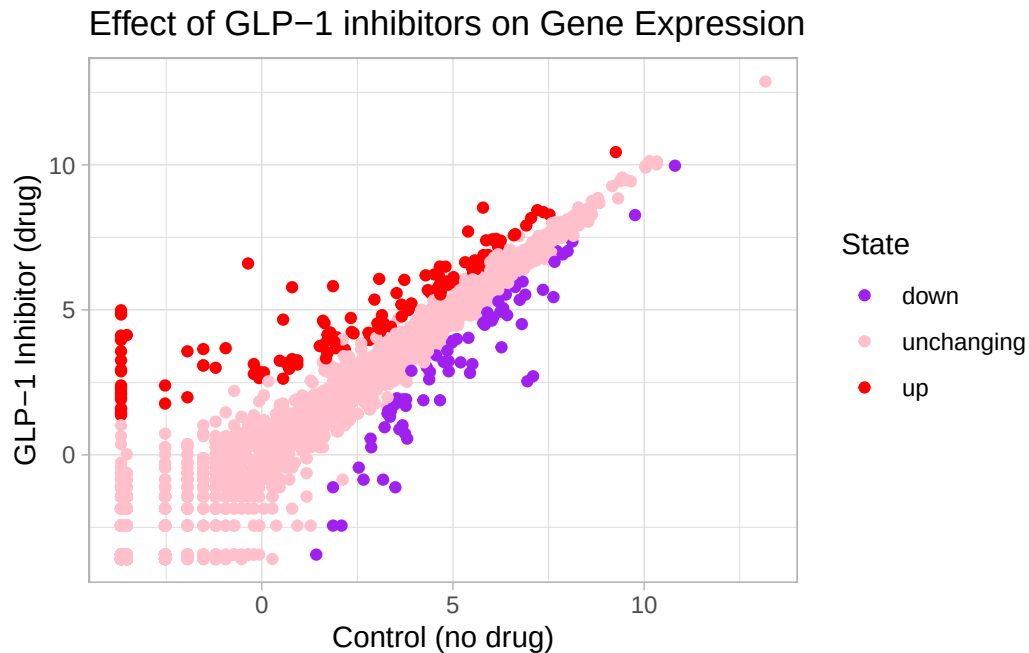
```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Version 1 plot = start simple by getting some ink on the page.

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2, col=State) +
  geom_point() +
```

```
scale_color_manual(values = c("purple", "pink", "red")) +
labs(title = "Effect of GLP-1 inhibitors on Gene Expression", x = "Control (no drug)", y =
theme_light()
```



Let's color by State up, down or no change

```
genes$State
```

```
[1] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[6] "unchanging" "unchanging" "unchanging" "unchanging" "up"
[11] "up"         "unchanging" "unchanging" "unchanging" "unchanging"
[16] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[21] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[26] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[31] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[36] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[41] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[46] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[51] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[56] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[61] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
[66] "unchanging" "unchanging" "unchanging" "unchanging" "unchanging"
```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Going Further with gapminder

Here we explore the famous `gapminder` dataset with some custom plots.

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many rows does this dataset have?

```
nrow(gapminder)
```

```
[1] 1704
```

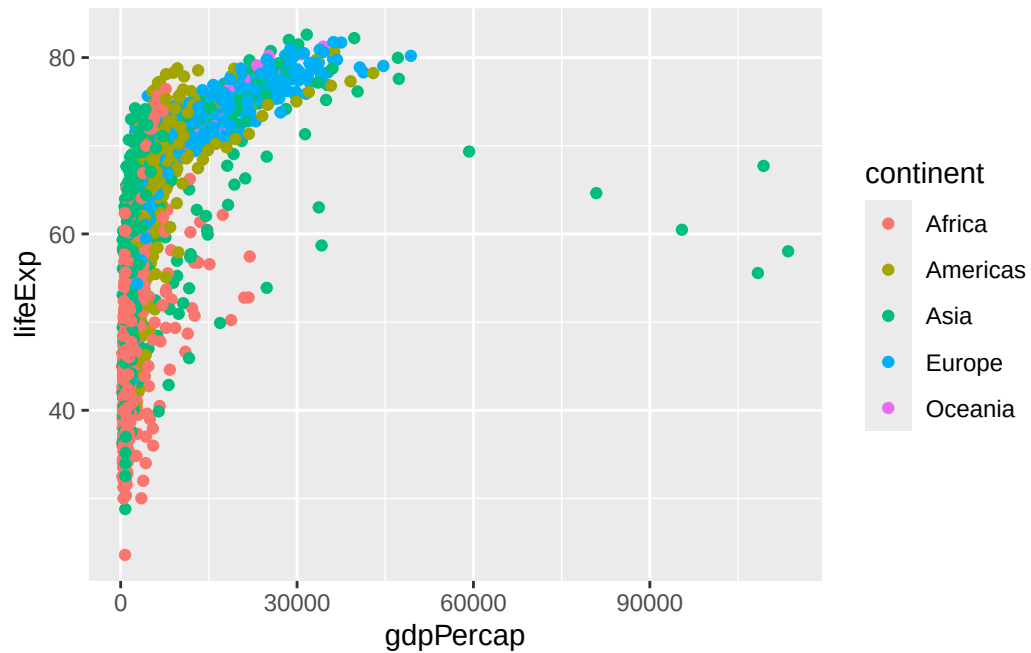
Q. How many different continents are in this dataset?

```
table(gapminder$continent)
```

Africa	Americas	Asia	Europe	Oceania
624	300	396	360	24

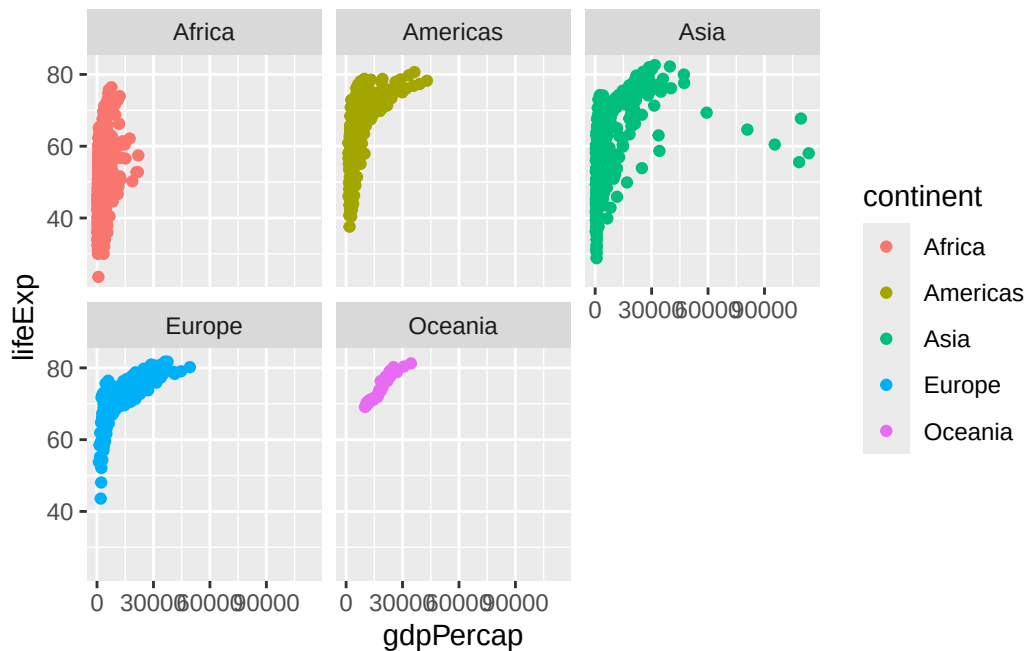
Version 1 plot GDP vs lifeExp for all rows

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col= continent) +
  geom_point()
```



I want to see a plot for each continent - in ggplot lingo this is called “faceting”

```
ggplot(gapminder) +  
  aes(gdpPercap, lifeExp, col= continent,) +  
  geom_point() +  
  facet_wrap(~continent)
```



First look at the dplyr package

Another add-on package with a function called `filter()` that we want to use.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
filter(gapminder, year == 2007, country == "Ireland")
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Ireland	Europe	2007	78.885	4109086	40676


```
input <- filter(gapminder, year == 2007 | year == 1977)
```

```
ggplot(input) +  
  aes(gdpPercap, lifeExp, col = continent) +  
  geom_point() +  
  facet_wrap(~year)
```

